

Raghav Verma

New Delhi, India | raghav.verma.work@gmail.com | GitHub | raghav-verma.com | LinkedIn

Summary

Full-Stack Engineer specializing in real-time data visualization and computer vision powered analytics. Experienced building high-performance React systems integrating structured ML outputs for interactive video analysis workflows.

Technical Skills

Languages	TypeScript, JavaScript (ES6+), Python, SQL
Frontend	React 18, Next.js 14, Vite, Tailwind CSS, shadcn/ui, HTML5, CSS3
Visualization	Three.js, React Three Fiber, WebGL, Canvas API
AI / ML & CV	YOLOv8, pose estimation, keypoint detection (COCO-17), motion analysis, biomechanical metrics, geometric feature extraction, signal smoothing, time-series inference pipelines
Backend	Node.js, REST APIs, PostgreSQL, Prisma ORM
Tools	Git, Docker, Postman, Jest, Vitest

Professional Experience

SDE Intern — **Khel.AI** Feb 2026 – Present

- Built real-time video analysis interfaces integrating pose keypoint outputs into React dashboards.
- Developed frame-accurate video tooling with synchronized overlays using Canvas API.
- Implemented geometric computations for joint angles and motion alignment in-browser.
- Optimized rendering performance for dense frame-level data across high-frequency video inputs.
- Designed reusable visualization components for high-frequency motion data.
- Improved metric stability using smoothing and outlier filtering techniques.
- Implemented modular component architecture supporting synchronized multi-video comparison workflows.
- Engineered efficient state synchronization logic for consistent frame-level metric updates.
- Structured visualization pipelines for time-series motion analytics.

Frontend Engineer (Contract) — **Hypeliv Solutions Pvt. Ltd** Aug 2025 – Jan 2026

- Improved Next.js performance reducing LCP from 2.5s to 1.5s via code splitting and lazy loading.
- Built real-time trading interfaces integrating WebSocket market streams.
- Defined type-safe API contracts with backend teams.
- Improved reliability of UI data handling through schema validation.

Frontend Engineer Intern — **The TechnoLabs** Jan 2024 – Jul 2024

- Built React invoice system with validation and PDF export.
- Integrated REST APIs for full invoice lifecycle.
- Automated OCR dataset validation using Python scripts.
- Improved data consistency checks using structured validation logic.

Open Source Projects

Meridian — **API Resilience SDK** npm | GitHub

- TypeScript SDK providing retry, fallback, and circuit breaker logic for external APIs.
- Supports OpenAI, Anthropic, Stripe, GitHub with unified response typing.
- Reduces integration complexity via consistent error handling patterns.
- Designed for ESM/CommonJS compatibility with zero runtime dependencies.

DevTrackr — **GitHub Signals SDK** Live Demo | GitHub

- Extracts and normalizes GitHub developer signals into structured JSON responses.
- Implements pagination and rate-limit handling with full TypeScript typings.
- Provides structured datasets suitable for analytics dashboards.

Education

Master of Computer Applications 2024 – 2026

Vivekananda Institute of Professional Studies, GGSIPU

Bachelor of Computer Applications 2021 – 2024

Bennett University